MagPress M4: what it costs to build in India

Prices researched 28–29 October 2026 from sellers' own pages, IndiaMART, TradeIndia, Moglix and institute rate cards. All prices are before GST (18 %) and transport. Treat every figure as ±30 % and get two or three written quotes against the drawings. Andhra Pradesh and Delhi NCR are listed first; the last column names the best option anywhere.

Read this first: whose numbers are these?

- **Every MagPress figure in this sheet is our build cost:** what you would pay shops and sellers to make or buy each part of the MagPress. It is **not** the market price of a machine.
- **"All eight kits ₹1.6–3.2 lakh" is 8 × ₹20,000–40,000:** the cost of having every MagPress size kit made (4 square + 4 round). Nobody sells MagPress kits, so there is no market price for them. Page 2 shows why one kit costs ₹20,000–40,000.
- **You do not need all eight.** Start with one kit (₹20,000–40,000) and add a size only when you will sell it.
- **Market prices, for comparison:** a ready-made Indian square magnet press makes **one** size and costs ₹10,000 (50 × 50, plastic body, with cutter and 100 sets) to ₹30,000 (50 × 50, metal body), and about ₹46,000 for 80 × 53 with a cutter.

Status: square size kits on hold

The frame, drive and fits passed verification. The square kits are on hold until your findings from taking apart a bought magnet are in, so the cost of a square kit and its back consumable may still change. The frame, universal die, bought parts and round kits are priced as drawn (Rev E).

1. Totals (our build cost)

Block	What it covers	Approximate cost
Frame and drive	About 40 machine hours (base, rails, keepers, stop blocks, die table, columns, crown, ram, clevis, holder, cheeks, lever, links, pins, rods, spacers, pilot, post), profile cutting, about 85 kg of steel, aluminium and bronze, hardening of pins, black oxide, anodise, powder coat	₹40,000 – 60,000
Universal die	About 17 machine hours (die body, shank, upper carrier, two lower carriers, two station bases), material, black oxide	₹12,000 – 20,000
Bought parts	Springs, bushings, fasteners, pins, plungers, detent pin, small parts (section 3)	₹12,000 – 25,000
Press without a kit		**₹65,000 – 1,05,000**
One size kit	Six hardened forming parts and their small parts (page 2)	₹20,000 – 40,000
Press with one kit	**What you need to start making magnets. About ₹1.7 lakh if every part is made at an institute tool room's rates**	**₹0.85 – 1.45 lakh**
Each extra kit	Only for sizes you will sell	₹20,000 – 40,000
All eight kits	8 × one kit. Our cost, not a market price	₹1.6 – 3.2 lakh
Press with all eight kits	Press + 8 kits	₹2.25 – 4.25 lakh
Forming trial (recommended before hardened kits)	One size's six forming parts in 7075 aluminium or soft D2, pressed in a 1–2 t bench arbor press, to prove the folds on real tin, paper and film	₹15,000 – 25,000
Commercial consumables to measure	One pack of bought square-magnet sets, measured with calipers, to fix the reworked square kit's sizes	₹300 – 500

Market prices for comparison (ready-made)

Item	Where	Price
Square magnet press 50 × 50, plastic body, cutter and 100 material sets	Copier Bazar (+91 88899 15080)	₹10,000
Square magnet press 50 × 50, metal body	IndiaMART listings; Copier Bazar	₹16,500 – 30,000
Square magnet press 80 × 53, with cutter	Copier Bazar	₹46,000
Bought material sets (sheet, film, back)	Same sellers	₹25 – 27 per magnet

For one or two square sizes, a bought press is cheaper. The MagPress is worth building for several sizes on one heavier frame, for round and square on the same press, and for cheaper consumables (≈ ₹10–15 per magnet against ₹25–27).

2. Why one size kit costs ₹20,000 – 40,000

A size kit is six precision forming parts in hardened tool steel: the plug, upper ring, pickup ring A, pickup pedestal, crimp ring B and crimp anvil. They are cut to 0.01 mm by wire EDM *after* hardening, so they keep their size, and their forming edges are polished. That is what a press die costs in any tool room. Most of the price is machine time, not steel.

Step	Basis	Cost per kit
D2 tool steel ("HCHCr")	About 8 kg bought (6 kg finished) at ₹260–290 per kg	₹2,100 – 2,300
Soft CNC machining before hardening	About 5 h at ₹250–450/h (private shop) to ₹1,030–1,400/h (institute tool room)	₹1,300 – 7,000
Vacuum hardening, double temper, 58–60 HRC	Usually a minimum batch charge, so several kits together cost little more than one	₹2,000 – 5,000
Wire EDM after hardening	Openings and outlines of the six parts, one main and two skim cuts: about 60,000 mm² of cut face at ₹0.12–0.24 per mm²	₹7,200 – 14,400
Grinding, edge radii, lapping, polishing	About 6 h: faces ground flat, the anvil cove and pedestal edge radius hard-milled or EDM'd, forming edges lapped to Ra 0.2	₹2,500 – 8,000
Kit small parts	Dowel pins, four M3 ball plungers, plug stud	₹1,000 – 2,500
Sum	**Rounded up to ₹20,000 – 40,000 for set-up and minimum charges on a one-off job**	**≈ ₹16,000 – 40,000**

How to bring it down

- Make one kit first, and later kits only for sizes you will sell.
- Have several kits made together: one hardening batch and one EDM set-up are shared between them.
- Round kits are mostly turned and bored on a lathe, so they need much less wire EDM than square ones.
- Private shops charge a third to half of institute tool-room rates; use a tool room only for the hardening and wire EDM if needed.

3. Where to get each part

Machining

Work	Andhra Pradesh	Delhi NCR	Best option / price
Size-kit parts: CNC rough, wire EDM after hardening, grinding, CMM check	**MSME Technology Centre, Visakhapatnam**, Plot 6, Industrial Corridor, Pudi, Atchutapuram 531011; +91 95153 97533, 08924-282600; info@msmetcvizag.org. CNC mills and lathes, wire-cut and die-sink EDM, grinders, CMM. No vacuum furnace listed: send kit parts out for hardening	NSIC Technical Services Centre, Okhla (99799 33266, 011-26826801 ext. 228); private tool rooms in Okhla, Mayapuri, Wazirpur, Faridabad	**CITD Hyderabad** (MSME Tool Room, Balanagar), the nearest full tool room to AP: wire cut, EDM, VMC, CNC lathes, grinding. Call or visit
Frame, drive, universal die (C45, S355, 42CrMo4)	MSME TC Visakhapatnam; CNC shops in the Vijayawada and Visakhapatnam Autonagars	CNC job shops in Okhla, Mayapuri, Naraina, Faridabad	Any shop with a 3-axis VMC and a CNC lathe

Work	Andhra Pradesh	Delhi NCR	Best option / price
Laser or plasma cut blanks (base plate, lever); laser-cut backs	Local fabricators	Faridabad laser job work, ₹25–30 per minute for steel	Acrylic: ₹50–245 per sq ft, or ₹15–90 per running foot

Rates to check quotes against: VMC ₹250–450/h at small private shops, ₹1,030–1,400/h at institute tool rooms (CSIR-CMERI). CNC lathe ₹400/h (IIT Kanpur, industry) to ₹1,150/h (CMERI). Wire EDM ₹0.12–0.24 per mm² of cut face (≈ ₹77–155 per sq inch) at private shops; ₹480–580/h at IIT Madras, ₹900/h at IIT Kanpur, ₹1,240/h at CMERI.

Heat treatment and finishing

Process	Parts	Andhra Pradesh	Delhi NCR	Price
Vacuum hardening 58–60 HRC, double temper	Size-kit parts (D2), about 6 kg per kit	Hyderabad tool rooms and heat treaters (Balanagar, Jeedimetla)	**voestalpine High Performance Metals India**, C-19, Site C, UPSIDC Surajpur, Greater Noida 201306; 85888 91063	₹2,000 – 5,000 per batch
Through-hardening and grinding	Pins A, B, O (EN31), station pilot, die holder (EN19)	Local heat treaters	Local heat treaters; Maxwell Tools, Rajpura (vacuum)	₹1,000 – 2,000
Black oxide	Frame and drive steel parts	Local platers	Local platers	₹1,000 – 2,000
Hard anodise	Die table (6061); mask the pilot and dowel holes	Hyderabad / Chennai platers	Delhi listing ₹3 per sq inch	₹500 – 800
Hard chrome	Columns, only if not bought as chrome bar	—	Greater Noida listing ₹4 per sq inch	₹500 – 800
Powder coat	Base plate	Local	Local	₹500 – 1,000

Materials and bought parts

Item	Spec and quantity	Andhra Pradesh	Delhi NCR	Best option / price
Kit tool steel	D2 (1.2379, "HCHCr"), about 8 kg bought per kit. A2 is rarely stocked in India; D2 is the one to order	Through the tool room	Loha Mandi, Naraina (e.g. Goel Steel Emporium); voestalpine Greater Noida (Böhler K110 = D2)	₹260–290 per kg generic; branded more
Carbon and alloy steel	C45 (≈ EN8/EN9) and 42CrMo4 (≈ EN19) bar and plate, about 45 kg	Stockists in Vijayawada / Visakhapatnam	Loha Mandi, Naraina	₹65–90 per kg
Base plate steel	S355 (≈ IS 2062 E350), 20 mm, 480 × 490, about 37 kg	RINL / SAIL stockists	Naraina	₹51–60 per kg + cutting
Columns	Ø25 × 210 mm, 2 off: 100Cr6 hardened, ground, hard-chromed, or induction-hardened chrome bar ground to Ø25 h6	—	Indian Hydraulic Products, Delhi: IH chrome bar ₹150 per kg (≈ ₹600 per metre at Ø25)	₹600–1,200 for both. Hydraulic rod is usually f7: ask for h6
Die table	6061-T6 plate, 12 mm, 340 × 150	Aluminium stockists	Aluminium stockists	₹350–450 per kg (≈ ₹700)
Bronze	CuSn8 / phosphor bronze rod, pivot spacers	Local	Local	₹400–1,000 per kg

Item	Spec and quantity	Andhra Pradesh	Delhi NCR	Best option / price
Die springs	ISO 10243: 8 × Ø10 × 32 green, 8 × Ø12.5 × 44 red	Order online	Pi-Tech Services, New Delhi; Standard Ring Pvt Ltd, Karnal Road	Prosoc and Maxima (Bengaluru), FIBRO India (Pune): ₹100–300 each, **₹2,000–5,000** for 16
PTFE-composite bushes (DU type)	16×18×15 (2), 16×18×20 (1), 12×14×20 (2), 12×14×6 (4)	Bearing shops	Shraddhanand Marg / G.B. Road bearing stores (e.g. RM Overseas, Nitin Machinery Stores)	₹15–150 each; ≈ ₹1,000
Bronze flanged bushes	Self-lubricating, 25 × 32 × 33, 2 off	Bearing shops	Same	₹300–600 each
Socket screws, nuts, set screws	Grade 12.9 (Unbrako or TVS), full list in the parts list	Local hardware	**Bazar Sirkiwalan, Chandni Chowk** (e.g. Reliance Fasteners, Panchsheel Fasteners); Moglix online	₹1,500–3,000
Shoulder screws ISO 7379	Ø6 × 10 (4), Ø6 × 12 (8)	Online	Unbrako dealers	₹80–200 each
Dowel pins, hardened m6	Frame and universal die about 13; per kit 6	Online	Unbrako dealers	₹500–1,000
Ball plungers	M3 × 4 per kit; M6 × 1	Online	MISUMI India (from ₹953); IndiaMART	₹1,000–4,000
Quick-release detent pin	Ø6, 64 usable, ring handle	Online	Jergens India (₹3,000) or generic stainless (₹200)	₹200–3,000
Return springs	Music wire Ø3.5, OD 31.5, free length 128, 2 off (made to order)	Local spring maker	Spring makers in Wazirpur / Karnal Road	₹300–800 for two
Nord-Lock washers	M8 (4), M16 (2)	Online	Online	₹1,500–3,000; or Loctite 243 with plain hardened washers
Small parts	Retaining rings, thumb screws, rubber pads, bumpers, push knobs, grip, POM sleeves, shim pack	Local	Local	₹1,000–2,000

4. Consumables, per magnet (63.5 × 63.5)

Item	Spec	Where	Per magnet
Tin blank	0.20 mm soft (T2) electrolytic tinplate, corners clipped	Optimum Steels, New Delhi, ETP sheet ₹72 per kg; Diamond Tin & Metal, Kolkata, 0.20 mm ₹97 per kg. Cut with a hand guillotine, or have a local press shop make a ₹5,000–15,000 blanking die per size	≈ ₹1
Photo	90–120 g/m² inkjet matte paper	Any stationer	≈ ₹1
Film	0.075 mm PET (Mylar)	Polyester film rolls ₹100–180 per kg; Sudershan Paper Mart, Delhi	< ₹1
Back	As drawn: ABS or acrylic 1.2 mm, laser cut. After the square-kit rework, probably a thick back that fills the shell (card or board about 2–4 mm), as in bought sets	Laser cutters; card and board from paper merchants (Chawri Bazar, Delhi, is the main paper market)	₹2 – 6
Magnet sheet	Self-adhesive flexible magnet, 0.5 mm, A4	IndiaMART sellers, ₹15–64 per A4 sheet (self-adhesive ₹64 in Greater Noida)	₹4 – 7
Total		**Against ₹25–27 for bought material sets**	**≈ ₹10 – 15**

5. How to order

1. Wait for the square-kit rework if you want square magnets; the frame, universal die and round kits do not depend on it, but order everything together once the design is settled.
2. Send the drawing PDF and DXF zip (and the STEP files if the shop has CAM) with the material and heat-treatment column of the parts list. Ask for a price per part and an inspection report on the H7 bores, the 210.00 hole pitch (base, crown, ram) and the kit openings.
3. Kit parts: rough machine, vacuum harden, then wire EDM and grind, in that order. Ask the tool room to arrange the hardening if it has no vacuum furnace.
4. Buy the springs, bushings, pins and fasteners yourself before the parts arrive, so the shop can test-fit them.
5. Start with one size kit; order more after the first magnets come out right.

Sources

1. MSME Technology Centre Visakhapatnam, machines and contact: https://www.msmetcvizag.org/list-of-machines/ · https://www.msmetcvizag.org/contact-us/
2. NSIC Technical Services Centre, Okhla: https://nsic.co.in/NTSC/Okhla
3. Faridabad precision machining and laser cutting: https://dir.indiamart.com/faridabad/precision-machining-services.html · https://dir.indiamart.com/faridabad/laser-cutting-services.html
4. Acrylic cutting rates: https://m.indiamart.com/impcat/acrylic-cutting-service.html
5. VMC job work: https://m.indiamart.com/vanshi-engineering · CSIR-CMERI rate chart: https://cmeri.res.in/sites/default/files/Brochure-MSME%20Rate%20Chart.pdf
6. IIT Kanpur charges: https://www.iitk.ac.in/mse/MSE-Facilities/MTL/charges.php · IIT Madras wire EDM: https://cstf.iitm.ac.in/downloads/Guidlines%20to%20use%20wire%20EDM%20machine.pdf · Wire cutting rates: https://m.indiamart.com/impcat/wire-cutting-services.html
7. voestalpine India: https://www.voestalpine.com/highperformancemetals/india/en/contact-us/
8. Hard anodising: https://m.indiamart.com/impcat/hard-coat-anodizing.html · Hard chrome: https://m.indiamart.com/impcat/hard-chrome-plating.html
9. D2 tool steel: https://www.tradeindia.com/products/d2-tool-steel-c3026899.html · Die steels: https://m.indiamart.com/impcat/die-steels.html
10. EN8 price benchmark: https://www.bigmint.co/intel/detail/india-alloy-steel-round-bar-prices-remained-stable-w-o-w-35171 · Delhi bright bar: https://m.indiamart.com/bkengineeringtools/bright-steel-bars.html · MS plates: https://m.indiamart.com/impcat/ms-plates.html
11. Induction-hardened chrome bar, Delhi: https://www.aajjo.com/product/induction-hardened-rod-in-delhi-indian-hydraulic-products · Phosphor bronze: https://www.tradeindia.com/products/phosphor-bronze-rods-1685517.html
12. Die springs: https://m.indiamart.com/pi-techservices/die-spring.html · https://www.tradeindia.com/products/die-springs-563445.html · MISUMI: https://in.misumi-ec.com/vona2/maker/misumi/mold/K1000000000/K1001000000/K1001020000/
13. Bushings: https://m.indiamart.com/impcat/dry-bushings.html · https://m.indiamart.com/impcat/bronze-bushes.html · https://m.indiamart.com/rm-overseas · https://m.indiamart.com/nitin-machinery-stores
14. Fasteners: https://m.indiamart.com/reliancefasteners/profile.html · https://m.indiamart.com/panchsheelfasteners/profile.html · https://www.moglix.com/unbrako-m5×10mm-full-threaded-socket-head-cap-screw-122243-pack-of-500/mp/msniot8tjc1
15. Ball plungers: https://in.misumi-ec.com/vona2/detail/110310869849 · Ball lock pins: https://m.indiamart.com/impcat/ball-lock-pins.html
16. Tinplate: https://m.indiamart.com/optimum-steels · https://m.indiamart.com/diamond-tin-metal · Polyester film: https://m.indiamart.com/impcat/polyester-film.html · Magnetic sheets: https://m.indiamart.com/impcat/magnetic-sheets.html
17. Ready-made presses: https://copierbazar.com/fridge-magnet-machines/ · https://www.indiamart.com/proddetail/fridge-magnet-making-machine-50x-50-mm-sqaure-2855464652830.html · https://www.indiamart.com/proddetail/50mm-square-fridge-magnet-badge-machine-kit-285760645

8055.html